PRACTICAL -2

1.Aim:

Use Telegram to control Raspberry Pi GPIO.

2. Prior Concepts:

i) Python Programming

ii) MQTT Protocol

3. New Concepts:

Controlling LED through telegram bot and Raspi

4. Hardware Required:

i)LED

ii) Breadboard

iii) Resistor

iv) Jumper wires

5. Software Required:

On Mobile Phone: Telegram:

Steps:

Download Telegram from playstore on your android phone.

i) Install Telegram.

ii) Open Telegram app in your system or mobile

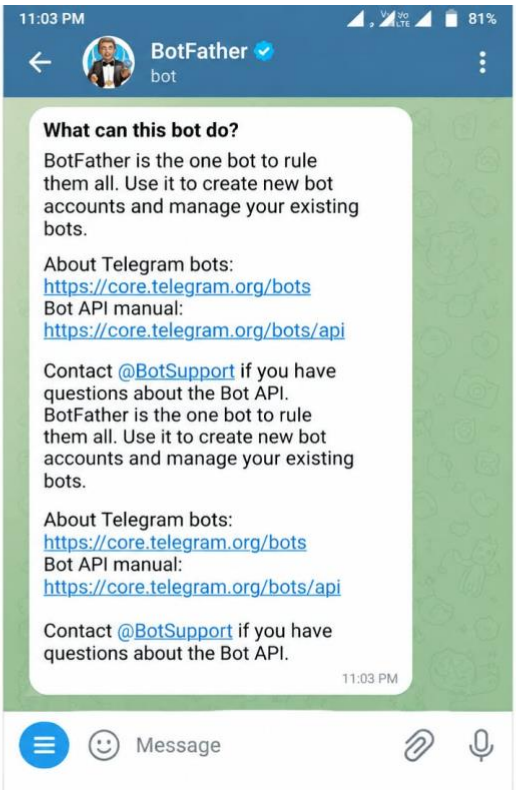
iii) Click On Start Messaging Button

iv) Enter your mobile number to register with telegram service.

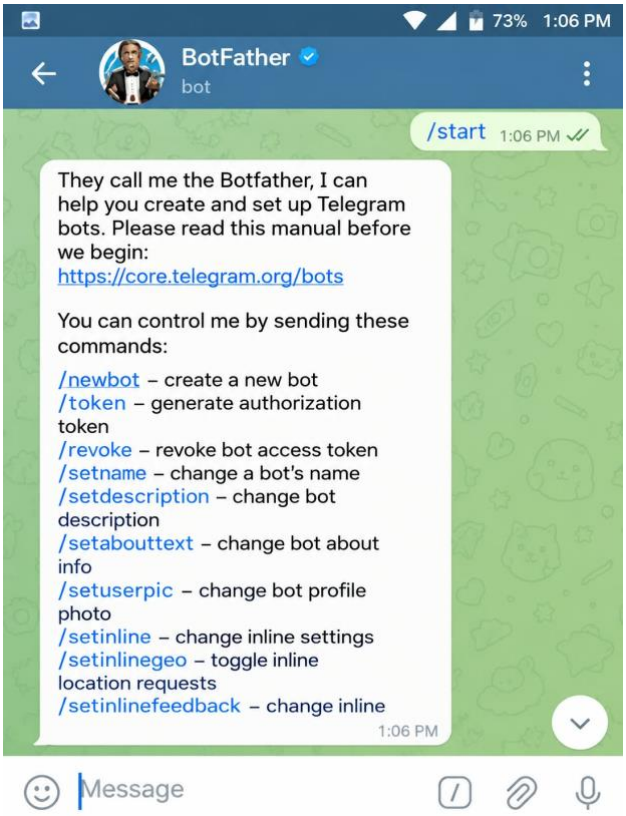
v)Search for name “BotFather”.



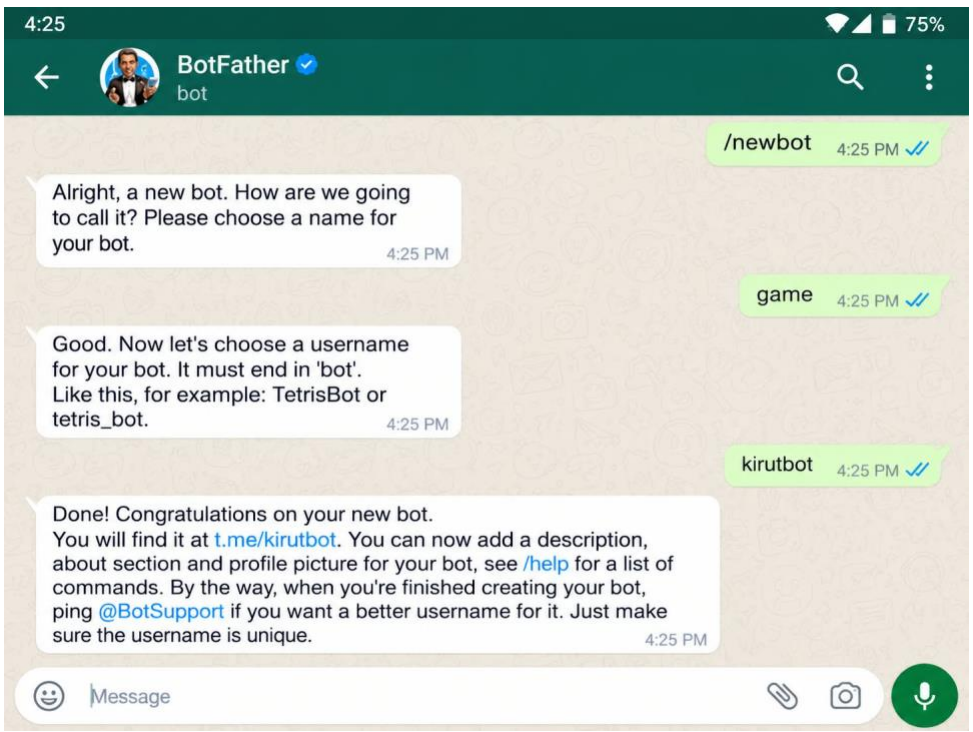
vi) Click on ”BotFather”



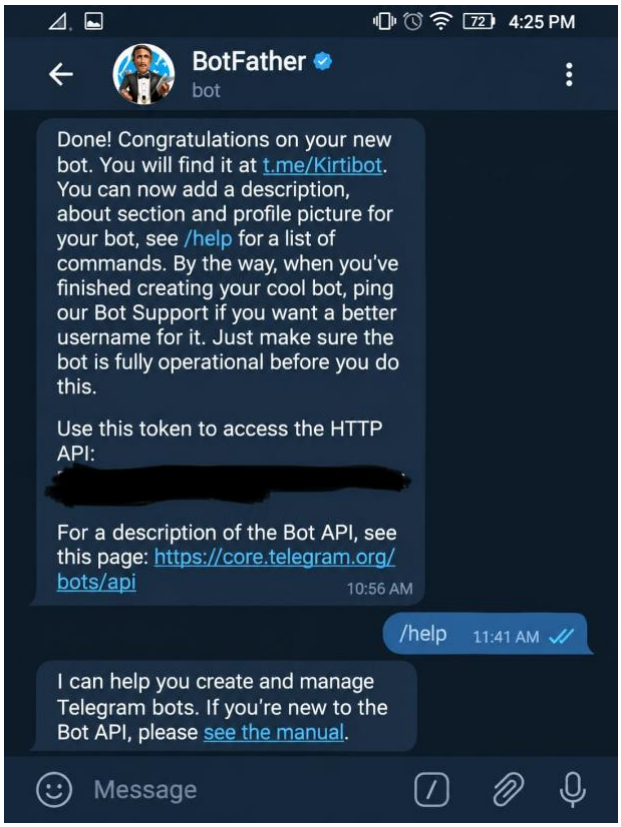
vii)To start “BotFather” type /start in message.



viii) Now type /newbot in message and then give name to your BOT and Username also:



ix) Obtain access token:



This same token we are supposed to use in our code in raspberry pi to connect to our BOT.

6.Procedure

Set up On Raspberry Pi:

1. Install "Python Package Index" and Telepot using:

python3 -m pip config set global.break-system-packages true

sudo apt install python3-pip

pip3 install telepot

2.Test your bot using python3 IDLE and type:

import telepot

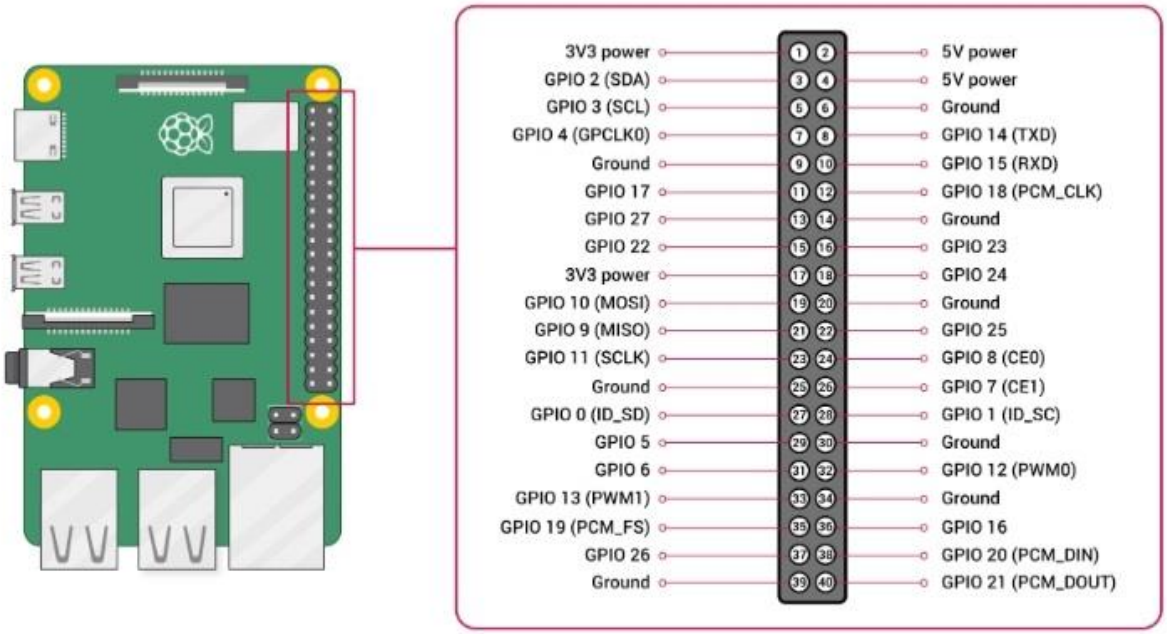
bot = telepot.Bot('Bot Token')

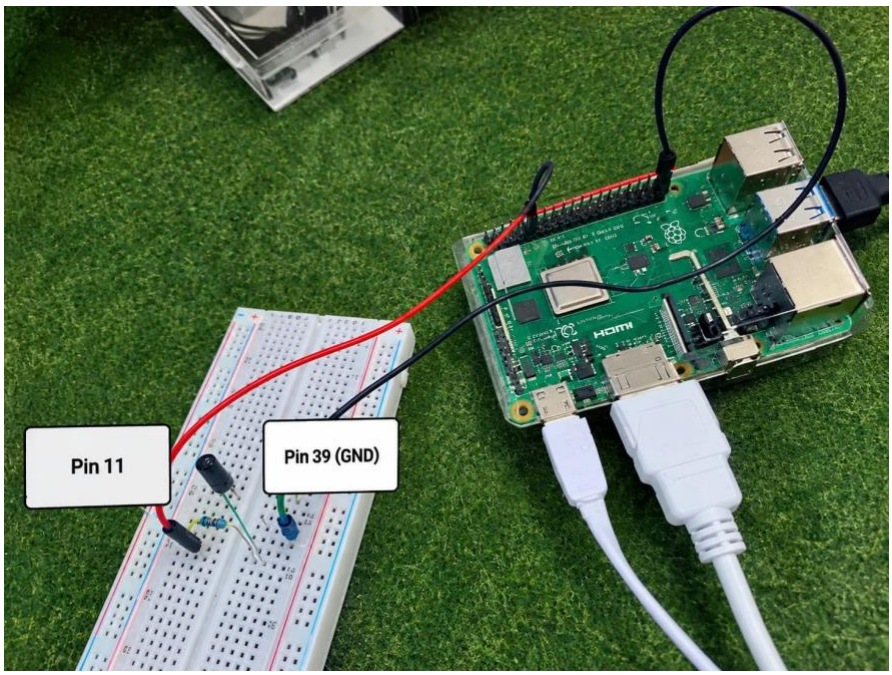
bot.getMe()

If it prints your bot details means everything is correct.

If not then token is wrong.ii. Connection of LED to Raspi.:

- Connect LEDs on breadboard
- Connect resistor to one end of led (to Positive end or anode).
- Connect other end of resistor to GPIO pin on raspberry Pi.
- Connect other end of LED to ground (negative end or cathode).





7. Implementation with Python Code

Question 1: Write a program to control single LED

Code:

```
import sys
import time
import random
import datetime
import telepot
import RPi.GPIO as GPIO
GPIO.setmode(GPIO.BOARD)
GPIO.setup(11, GPIO.OUT)
def handle(msg):
chat_id = msg['chat'] ['id']
command = msg['text']
try:
print 'Got command:', command
if command == 'on':
bot.sendMessage(chat_id, "LED on")
GPIO.output(11,GPIO.HIGH)
elif command =='off':
bot.sendMessage(chat_id, "LED off")
GPIO.output(11,GPIO.LOW)
elif command == 'stop':
exit()
bot = telepot.Bot('Bot Token')
bot.message_loop(handle)
print 'I am listening...'
```

```
while 1:
        time.sleep(10)
except TelegramError:
print ‘’
```

Note: do not forget to replace 'Bot Token' with your token received from telegram

8. Results:

a) For Single LED

